Differential-protection probabilistic scoring for methylome-guided ThermoCas9 target-site ranking

Technical memo from the thermocas methylome-analysis framework

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Abstract

Methylation-sensitive Cas9 variants target genomic loci that are hypomethylated in disease cells and methylated in matched normal cells. Selecting such loci from genome-scale array data is a ranking problem whose scoring axis must be cohort-agnostic and must report its own uncertainty. A first-pass probabilistic composite `p_targ × p_prot × p_trust` failed empirically — `p_prot` encoded a static "normal is methylated above 0.5" assumption that is anti-predictive on bulk normal comparators (AUC 0.38). We replace it with a differential factor `p_diff = P(β_normal − β_tumor > δ)` under an independent-normal approximation on per-probe summaries. On the single independent primary endpoint — AUC at the validated target probes from Roth et al. (2026) Fig. 5d on Roth's own MCF-7/MCF-10A EPIC v2 cohort (GSE322563) — the new composite (V2.5) reaches 0.990, compared with 0.821 for the deterministic V1 score and 0.928 for the deprecated V2 `tumor_only` mode. A second matched MCF-7/MCF-10A cohort (GSE77348) is the development cohort on which the differential margin δ was tuned and is reported as supporting evidence, not as an independent confirmation. Every benchmark result emits `tie_band_size_at_k` and `precision_at_k_{min,max}` so that low-replicate top-K numbers are reported as adversarial intervals, not point estimates. A separate cross-series run at the Sanger GDSC breast panel is documented as an out-of-distribution label-transport boundary case, not a generalization test, because Sanger's MCF-7 is methylated at the sites where Roth's

MCF-7 is unmethylated. V1 remains the stable-release default; V2.5 is the recommended probabilistic mode for matched cell-line cohorts; all code and benchmark artifacts are public.

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1 · Background

1.1 ThermoCas9 is methylation-sensitive at the PAM

Roth et al. (2026) characterized *Geobacillus thermodenitrificans* T12 Cas9 (ThermoCas9) biochemically, structurally, and in human cell-line editing assays. The central result: cleavage efficiency is governed by the methylation state of the fifth position of the PAM (`5'-NNNN**C**GA-3'` or `5'-NNNN**C**CA-3'`), not by methylation within the protospacer. Methylated 5-methylcytosine (5mC) at the PAM abolishes binding; unmethylated PAMs are cleaved as usual. Cryo-EM structures at 2.2–2.8 Å resolved the protein contacts that discriminate methylated from unmethylated cytosine.

The practical consequence is that ThermoCas9 gives a natural mechanism to selectively edit in cells where a target locus is hypomethylated while sparing cells where the same locus is methylated. Roth demonstrated this in MCF-7 vs. MCF-10A at three breast-cancer loci: *EGFLAM* (a control target), *ESR1*, and *GATA3*. The underlying methylation of the PAM cytosine at those loci was profiled on the Illumina Infinium MethylationEPIC v2 array and reported in Supplementary Figure 5d of the main paper alongside the editing results.

1.2 Target-site selection is a ranking problem

Given a reference genome, the ThermoCas9-compatible PAM sites on any chromosome number in the millions. To pick candidates for wet-lab follow-up, one wants a ranked list where the top scorers are *biologically coherent* — meaning the candidate's PAM cytosine is unmethylated in the disease cohort and methylated in the matched normal — *and* where the confidence of the underlying methylation measurement is part of the score. This is a classical ranking-with-covariates problem, and the natural scoring target is some decomposition of a probability of "therapeutic selectivity."

This memo describes the specific decomposition we ended up shipping after two earlier attempts failed on empirical cohorts.

1.3 Notation

For one candidate site on one cohort we carry five per-probe summary statistics from the cohort's methylation array: `β_tumor_mean`, `β_tumor_q25`, `β_tumor_q75`, `β_normal_mean`, `β_normal_q25`, `β_normal_q75`, plus per-side sample counts `n_tumor` and `n_normal` and an `EvidenceClass` capturing the distance between the candidate's PAM cytosine and the nearest assayed CpG probe (`EXACT`/`PROXIMAL_CLOSE`/`PROXIMAL`/`REGIONAL`/`UNOBSERVED`).

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2 · The V2 misspecification

2.1 First-pass composite

The first-pass composite followed a standard decomposition:

```
p_therapeutic_selectivity = p_targ × p_prot × p_trust
```

- `p_targ = P(β_tumor < 0.30)` — probability that the tumor's PAM cytosine is unmethylated enough for ThermoCas9 to cleave. Estimated via a method-of-moments Beta fit to `(mean, IQR/1.349)` with the regularized incomplete beta computed in pure stdlib via Lentz's continued fraction; falls back to a piecewise-linear CDF when the Beta moments are ill-defined.

- `p_prot = P(β_normal > 0.50)` — probability that the normal's PAM cytosine is methylated enough to protect the normal cell.
- `p_trust` — confidence factor that scales with `EvidenceClass` (0.95 at EXACT, 0.15 at REGIONAL, 0.0 at UNOBSERVED) and saturates linearly in `min(n_tumor, n_normal)` up to `ramp_n = 30`.

2.2 Empirical failure of `p_prot`

We ran a factor ablation on a structural surrogate for the Roth MCF-7 vs. MCF-10A comparison: GEO series GSE77348, an independent 2016 breast cell-line methylation series on HM450 (not the Roth samples). With 1,687 gene-region-filtered "positives" at the Roth genes, the measured AUCs were:

axis	AUC (loose)	AUC (tight)
`v2_tumor_only` (p_targ × p_trust)	0.733	0.770
`p_targ` alone	0.683	0.717
V1 `final_score`	0.657	0.628
`naive_selectivity` (β_n − β_t)	0.608	0.571
`p_targ × p_prot`	0.503	0.488
`p_prot` alone	**0.384**	**0.343**

`p_prot` is anti-predictive. Multiplying it against `p_targ` destroys the signal. The cause is the semantics: `P(β_normal > 0.5)` assumes that the normal comparator *methylates the target*. That assumption holds for Roth's specific MCF-10A cell line at the three *ESR1* / *GATA3* / *EGFLAM* probes they chose to validate (genes that are silenced in non-tumorigenic mammary epithelium) but does not generalize to bulk adjacent-normal cohorts or even to MCF-10A at gene-body probes.

2.3 V2.4 — keep the composite, drop `p_prot` by default

The intermediate fix (release v0.4.0) kept the composite but added a `probabilistic_mode` cohort YAML key so users could explicitly opt in or out of the anti-predictive factor. Default changed to `tumor_only = p_targ × p_trust`. This improved global AUC but the top-100 collapsed onto "always-unmethylated" loci (candidates where both tumor *and* normal are low-methylated) — good AUC, unusable for target discovery. The deeper fix was needed.

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3 · V2.5 · differential-protection probabilistic mode

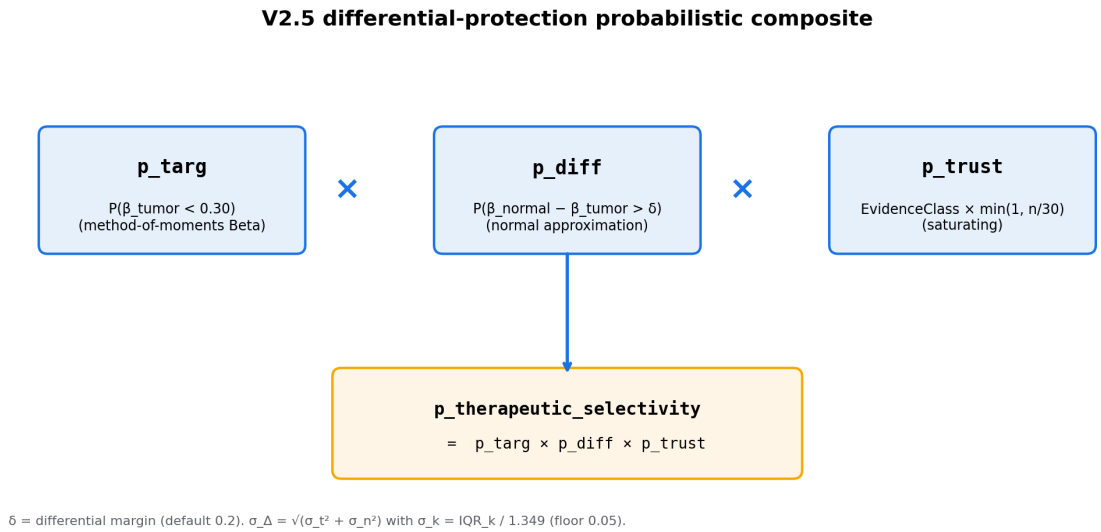

Figure 1: Figure 1 · V2.5 mode-formula schematic

Figure 1. The V2.5 composite. Three factors multiply: `p_targ` (tumor unmethylated at the PAM cytosine), `p_diff` (differential methylation gap exceeds δ under a normal approximation on per-probe β summaries), and `p_trust` (evidence-class confidence, saturating in min sample count). The result is `p_therapeutic_selectivity`, the stored scalar that downstream ranking consumes. The deprecated V2 `p_prot = P(β_normal > 0.5)` is replaced by `p_diff` to remove its static-threshold assumption about the normal side.

3.1 Formulation

Replace the threshold-based `p_prot` with a differential factor:

```
p_diff(δ) = P(β_normal − β_tumor > δ)
```

estimated under an independent-normal approximation:

```
σ_k   ≈ IQR_k / 1.349                (k ∈ {tumor, normal}; floor at 0.05)
σ_Δ²  = max(σ_t, floor)² + max(σ_n, floor)²
z     = (δ − (μ_n − μ_t)) / σ_Δ
p_diff = 1 − Φ(z) = 0.5 (1 − erf(z / √2))
```

with Φ the standard normal CDF. The composite becomes:

```
p_therapeutic_selectivity = p_targ × p_diff × p_trust
```

This makes no absolute-threshold claim about the normal side. It asks only whether the normal-vs-tumor gap exceeds δ in the sense of a normal-approximation confidence interval on the per-probe summaries.

3.2 Choice of δ

An offline δ sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 surrogate selected δ = 0.2 by joint optimization of AUC and `P@100`. Higher δ values monotonically reduce both — the factor becomes too strict to include candidates with moderate but real differentials. This default is exposed as `differential_delta` in the cohort YAML. We carry the δ used on every emitted `ProbabilisticScore` record so that reviewers can tell from a single JSONL row which margin was in effect.

3.3 Why a normal approximation instead of a difference-of-Betas

A proper Bayesian treatment would model β-values as Beta distributions and compute the exact difference distribution numerically. For the IQR ranges seen on HM450 / EPIC v2 arrays (~0.05–0.20), the normal approximation is close to the correct answer and has the advantage of being a two-line stdlib closed form. A σ floor of 0.05 prevents σ_Δ from collapsing to zero at boundary β-values (very common for CpG islands methylated at 0 or 1). These tradeoffs are documented in the code; a downstream user who wants the exact difference-of-Betas path can swap it in.

3.4 The low-n tied-band prediction

With the new factor isolated, we can predict before running any new cohort. `p_trust` is `EvidenceClass.base × min(1, min(n_t, n_n) / 30)`: piecewise-linear in min sample count up to `ramp_n = 30`, then a constant ceiling per evidence class. It is **never** continuous-valued in the real-line sense — at n ≥ 30 it saturates to a discrete-by- EvidenceClass value (0.95 at EXACT, 0.75 at PROXIMAL_CLOSE, etc.). What changes between the low-n and high-n regimes is which factor *dominates* the composite ordering:

- **Low n (n ≪ 30).** `p_trust` scales every same-evidence-class candidate by the same uniform fraction `(n / 30)`. Per-side IQRs are wide (or zero, hitting the `σ_floor`), pushing both `p_targ` and `p_diff` toward saturation (≈1) for any candidate with clean tumor hypomethylation and a large normal/tumor gap. The composite is effectively `1 × 1 × p_trust` for that whole subset, producing a large tied band at the top of the score distribution.
- **High n (n ≥ 30).** `p_trust` is constant per EvidenceClass. Per-side IQRs are realistic (HM450 ranges of 0.05–0.20). The composite ordering is now driven by the continuous variation in `p_targ × p_diff`, with `p_trust` only multiplying by a per- evidence-class constant. The previously-tied subset spreads out by its underlying `p_targ × p_diff` values.

The falsifiable prediction: the K = 100 tied band on V2.5 should be large at n=2/3 per side and shrink to near-1 at n ≳ 30 per side. §5.1 and §5.3 test this directly across cohorts ranging from n=2/2 to n=305/50.

---

4 · Benchmark methodology

4.0 Pre-registration: primary endpoint, sensitivity, boundary

To guard against post-hoc optimization, the analysis design separates tuned components from evaluation. The hyperparameter `differential_delta` was tuned on a specific cohort, and that cohort is therefore *not* part of the protected primary endpoint.

- **Tuned, fixed before cross-cohort evaluation.** `differential_delta = 0.2` was selected by an offline sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 MCF-7/MCF-10A surrogate against the *pre-repair* (gene-universe) positives. The value was fixed in code before any other cohort was ingested or benchmarked. GSE77348 is the **development cohort** for δ; results on GSE77348 are tuned-on supporting evidence, not independent confirmation. The label-repair exercise that produced the Roth-validated positives is data preparation, not scoring-axis tuning, and it post-dates the δ choice.
- **Primary endpoint.** AUC at the `positives_roth_validated.txt` label set (n = 3 Roth Fig. 5d target probes, hg38 → hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples). This is the single independent paper-comparable test. δ was not tuned against this cohort; the label set was not derived from it.
- **Supporting evidence** (§5.1, §5.2). AUC on GSE77348 (the development cohort for δ, MCF-7/MCF-10A on HM450). Same matched cell-line biology as GSE322563 but profiled by a different lab on a different

platform, so cross-platform agreement on the V2.5 ordering is informative — but it is *not* an independent cohort under the pre-registration discipline because δ was selected on it.

- **Sensitivity analyses** (§5.2). The `narrow` (±50 bp) and `wide` (±500 bp) label sets are alternative label granularities; the cross-granularity AUC ordering stability is reported without retuning δ. `P@K` intervals on tied bands are reported but are expected to be uninformative on low-`n` cohorts (§3.4).
- **Tissue-cohort behavior** (§5.3). GSE69914 (n = 305 / 50 primary breast tissue) is *not* a paper-comparable test of Roth biology — it probes whether the V2.5 architecture's low-`n` tied-band behavior dissolves at high `n`, as predicted in §3.4.
- **Out-of-distribution boundary case** (§5.4). GSE68379 (Sanger GDSC breast panel × GSE69914 healthy normal) is *not* a generalization cohort; it is a label-transportability test. It is reported separately and is not pooled into any primary-endpoint average.

4.1 Cohorts

Four cohorts, all public HM450 or EPIC v2 methylation arrays:

cohort	regime	n tumor / normal	platform	role
GSE322563	Roth actual samples	2 / 2	EPIC v2	paper-comparable biology
GSE77348	MCF-7/ MCF-10A surrogate (DNMT3B paper)	3 / 3	HM450	δ development cohort (tuned-on supporting evidence)
GSE69914	primary breast tissue	305 / 50	HM450	high-n tissue validation
GSE68379	Sanger GDSC cell-line panel × GSE69914 healthy normal	52 / 50	HM450	orthogonal (see §5.4)

Ingest details:

- **GSE322563.** Supplementary β matrix `GSE322563_beta_matrix_EPIC_v2.txt.gz`. EPIC v2 probe IDs carry `_BC##` / `_TC##` / `_TO##` / `_BO##` beadchip suffixes; we strip to the canonical `cg*` identifier and intersect with HM450 — 80.7% retention. Roth-positive gene retention: ESR1 96%, EGFLAM 91%, VEGFA 90%, GATA3 83%. The main paper's data-availability statement prints `GSE32256`, which resolves to a *Paramecium* transcriptome series; the supplement's reporting summary correctly prints `GSE322563`. We verified the liftover by cross-checking our per-probe β values at the three Roth-Fig.-5d target positions against the paper's published β — every value agrees to two decimal places (EGFLAM 0.01/0.49; ESR1 0.07/0.94; GATA3 0.02/0.31).
- **GSE77348.** Per-sample β values computed from Unmethylated / Methylated intensities (β = M / (M + U + 100)); restricted to untreated replicates.

- **GSE69914.** Full SeriesMatrix β table; filtered to `status == 2` (sporadic breast cancer, non-BRCA1) for tumor and `status == 0` (healthy donor, non-BRCA1) for normal. Tumor-adjacent normals and BRCA1 carrier arms excluded.
- **GSE68379.** Sanger processed matrix has integer row indices; we rebuild cg probe IDs via the Illumina GPL13534 manifest (485,577 data rows; row N → IlmnID N); filtered to 52 primary-site-breast cell lines. No in-study normal — paired cross-series with GSE69914 healthy normals.

All cohort builder scripts are in `scripts/build_*.py`. Every run produces `tumor_summary.tsv` + `normal_summary.tsv` + `PROVENANCE.md` in the shared `LocalSummaryBackend` format; the scorer is platform-agnostic thereafter.

4.2 Positives (label repair)

Our first benchmark used "gene-universe" positives: HM450 probes in the neighborhood of Roth's validated gene symbols (~970 tight, 1,687 loose, on chr5/6/10). A diagnostic on GSE322563 revealed this label set is noisy: only 23% of Roth-gene probes have $\beta_n - \beta_t > 0.2$ on the actual Roth samples. Most are gene-universe members, not validated target loci.

We rebuilt the positives from Roth Supplementary Fig. 5d which names three exact MCF-7 / MCF-10A validated target coordinates in hg38. These were lifted to hg19 via Ensembl REST `/map/human/GRCh38/.../GRCh37`:

target	gene	hg38	hg19 (lifted)	β shift (hg19 vs hg38)
EGFLAM T11	EGFLAM	chr5:38,258,842	chr5:38,258,944	+102 bp
ESR1 T17	ESR1	chr6:151,690,043	chr6:152,011,178	+321,135 bp
GATA3 T18	GATA3	chr10:8,045,425	chr10:8,087,388	+41,963 bp

Three positives files result, at increasing granularity:

- `positives_roth_validated.txt` (n = 3): closest NNNNCGA candidate per Roth target.
- `positives_roth_narrow.txt` (n = 28): NNNNCGA candidates within ±50 bp of each Roth target.
- `positives_roth_wide.txt` (n = 142): NNNNCGA candidates within ±500 bp of each Roth target.

4.3 Score axes

Three axes are benchmarked on every cohort:

- **V1** — `final_score = sequence × selectivity × confidence − heterogeneity_penalty − low_coverage_penalty`. Deterministic, continuous-valued, tie band always 1.
- **V2 `tumor_only`** — `p_targ × p_trust`. Retained for audit; default in v0.4.0.
- **V2.5 `tumor_plus_differential_protection`** — `p_targ × p_diff × p_trust` with $\delta = 0.2$.

4.4 Platform harmonization and catalog scope

Two facts constrain interpretation of the reported numbers and belong in the methodology, not as a footnote:

- **Platform.** GSE322563 was profiled on Illumina EPIC v2 (GPL33022, ~935K probes); GSE77348, GSE69914, and GSE68379 are on HM450 (GPL13534, ~485K probes). To use a single scorer across both platforms, we harmonize EPIC v2 probe IDs to the HM450 canonical form by stripping the EPIC v2 beadchip design suffixes (`_BC##`, `_TC##`, `_TO##`, `_BO##`) and intersecting with the HM450 universe. Retention: 80.7% of HM450 probes map. Per Roth- validated gene: ESR1 96%, EGFLAM 91%, VEGFA 90%, GATA3 83%. A

native EPIC v2 probe annotation (manifest ingest + catalog rebuild) would remove this caveat but is not shipped; every EPIC v2-origin per-probe β reported here passes through the HM450 intersect.

- **Catalog scope.** The candidate catalog is chr5 / chr6 / chr10 only, filtered to candidates within ±500 bp of any assayed HM450 probe (~2.98M NNNNCGA / NNNNCCA candidates). The three chromosomes were chosen because they carry the Roth Fig. 5d validated targets (EGFLAM on chr5, ESR1 on chr6, GATA3 on chr10) and because they provide a realistic but not whole-genome test bed. All AUC / `P@K` / tie_band numbers in §5 are on this scope. Extending to whole-genome hg19 is an infrastructure change, not a scientific one.

4.5 Tie-band handling

Earlier iterations of the benchmark sorted candidates by score alone, with input-order as the implicit tie-break. On cohorts with low n, the score distribution's top can sit inside a tied band of 190–11,000 records, making the reported `P@K` a function of serialization order rather than of the scorer.

The shipped benchmark contract:

1. Sort primary by score descending, secondary by `candidate_id` ascending (`evaluate_ranking`'s explicit `tie_break_policy = "candidate_id_asc"`).
2. Report `tie_band_size_at_k`: number of records tied at the K-th position's score. When > 1, top-K membership is partially determined by the secondary key.
3. Report `precision_at_k_min` and `precision_at_k_max` (plus recall counterparts) — adversarial tie-break bounds over the tied band. The worst-case pushes every tied-band positive out of top-K; the best-case pulls them in. When `tie_band_size_at_k == 1` both bounds collapse to the observed `precision_at_k`.

These three fields are on every emitted `BenchmarkResult` JSONL row.

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5 · Results

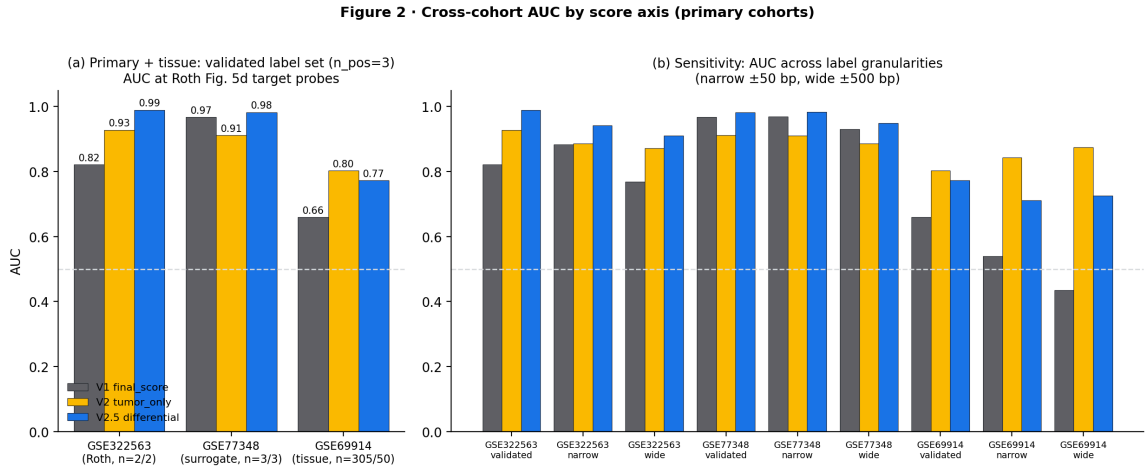

Figure 2: Figure 2 · Cross-cohort AUC by score axis

Figure 2. Cross-cohort AUC. **(a)** Validated-label AUC on each cohort. GSE322563 is the independent primary endpoint; GSE77348 is the development cohort on which δ was tuned (§4.0) and is shown as supporting evidence; GSE69914 is the tissue-behavior check for the §3.4 tied-band prediction. V2.5 is the highest-AUC axis on both matched cell-line cohorts; on tissue `tumor_only` takes a small AUC lead but at a tied band of 6,540 (§5.3 + Fig. 3). V1 falls below random on the tissue wide-label set in (b). **(b)** Sensitivity

over label granularity (validated ⟶ narrow ±50 bp ⟶ wide ±500 bp). The cohort-type × axis ordering is stable across granularities; only the magnitude varies. Dashed line at AUC = 0.5 (random). The out-of-distribution GSE68379 cohort is not included here — it is plotted separately (§5.4).

5.1 Primary endpoint — matched cell-line AUC at validated Roth probes

Primary endpoint (independent): AUC at `positives_roth_validated.txt` (n = 3 Roth Fig. 5d target probes, hg38 ⟶ hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples, EPIC v2). δ was not tuned on this cohort; the label set was not derived from it.

Supporting evidence (development cohort for δ): AUC on **GSE77348** (MCF-7 / MCF-10A on HM450, the DNMT3B-paper surrogate). Same matched cell-line biology, different platform and laboratory. Reported here because same-axis ordering on a second lab's MCF-7/MCF-10A is informative, but with the caveat that δ was selected on this cohort (§4.0) so its AUC is tuned-on, not independent.

axis	**GSE322563** (primary, n=2/2)	GSE77348 (δ-tuned, n=3/3)
V1 `final_score`	0.821	0.968
V2 `tumor_only`	0.928	0.912
V2.5 differential	**0.990**	**0.982**

On the independent primary endpoint (GSE322563 validated), V2.5 is the highest-AUC axis — +0.17 over V1 and +0.06 over the deprecated V2 `tumor_only` composite. On the development cohort (GSE77348 validated), V2.5 is similarly the highest-AUC axis (+0.01 over V1, +0.07 over `tumor_only`); this is consistent with the primary- endpoint result but is not an independent replication, since the same cohort selected δ = 0.2 pre-repair-labels (§4.0).

5.2 Sensitivity analyses: label granularity and P@K intervals

Label granularity (AUC). Stability of the primary-endpoint ordering under weaker label definitions (narrow ±50 bp, wide ±500 bp):

cohort	label set	V1	tumor_only	V2.5
GSE322563	validated (n=3)	0.821	0.928	**0.990**
GSE322563	narrow (n=28)	0.884	0.886	**0.942**
GSE322563	wide (n=142)	0.768	0.871	**0.910**
GSE77348	validated (n=3)	0.968	0.912	**0.982**
GSE77348	narrow (n=28)	0.969	0.911	**0.983**
GSE77348	wide (n=142)	0.931	0.887	**0.949**

V2.5 remains the highest-AUC axis on both cohorts at every label granularity. No retuning of `δ` between rows.

P@K intervals. On `n = 2 / 2` and `n = 3 / 3` cohorts the score distribution at K = 100 sits inside a tied band (§3.4); P@100 is reported as an adversarial interval `[min, max]` per §4.5. On GSE322563 narrow labels:

axis	P@100 observed	P@100 min	P@100 max	tie_band@100
V2.5 differential	0.000	0.000	0.020	190
tumor_only	0.000	0.000	0.020	10,005
V1 final_score	0.020	0.020	0.020	1

V2.5′s interval is [0.000, 0.020] — under the deterministic `candidate_id` ascending tie-break, zero narrow-positives land in top-100; under the best-case adversarial tie-break inside the 190-tied band, 2 of 28 could. V1′s interval collapses to the observed value because V1′s score is continuous and `tie_band = 1`. `tumor_only`'s tied band (10,005) spans two orders of magnitude more of the score distribution than V2.5′s; top-K on `tumor_only` is not a meaningful quantity on this cohort.

5.3 Tissue-cohort behavior — GSE69914 (high-n, tissue biology)

GSE69914 (n = 305 / 50 primary breast vs. healthy donor tissue) tests whether the low-n tied-band behavior predicted in §3.4 dissolves at $n \gg 30$:

label set	V1	tumor_only	V2.5	V2.5 tie_band@100
validated (n=3)	0.660	**0.803**	0.773	**2**
narrow (n=28)	0.539	**0.843**	0.711	2
wide (n=142)	0.435	**0.874**	0.726	2

Two findings:

1. **The tied-band prediction holds.** V2.5′s tied band at K = 100 shrinks from 190 at `n = 2 / 2` ⟶ 299 at `n = 3 / 3` ⟶ 2 at `n = 305 / 50`, with no hyperparameter changes between cohorts. At `n ≥ 30`, `p_trust` saturates at the EvidenceClass ceiling (0.95 for EXACT) — a per-class *constant*, not a continuous function of n. The tied band shrinks not because `p_trust` becomes continuous-valued but because `p_targ × p_diff` are no longer pushed to 1 by wide-IQR saturation and the σ floor; with realistic-width IQRs (0.05–0.20) the composite's ordering is driven by continuous variation in those two factors while `p_trust` just multiplies by the class constant.
2. **The mode ordering flips on tissue biology.** `tumor_only` wins AUC (+0.03 to +0.15 over V2.5), but `tumor_only`'s tied band is 6,540 at K = 100 — its top-K is not usable. V2.5 is the only probabilistic axis with an interpretable top-K on tissue. V1 collapses on tissue (AUC 0.44 on wide, below random).

5.4 Out-of-distribution boundary case — GSE68379

Reported here separately. ***Not*** *pooled with the primary-endpoint or sensitivity tables.*

A cross-series run against the Sanger GDSC 52-breast-line panel × GSE69914 healthy normal (n = 52 / 50) at the Roth-validated labels produced systematically inverted AUC:

label	V1	tumor_only	V2.5	V2.5 tie_band@100
validated	0.510	0.222	0.197	1,602
narrow	0.485	0.228	0.169	1,602
wide	0.407	0.462	0.269	1,602

The root cause is a probe-level biology mismatch. At the three Roth Fig. 5d target probes:

probe	gene	Roth MCF-7 β (GSE322563)	Sanger MCF-7 β (GSE68379)
cg05251676	EGFLAM	0.01	**0.92**
cg25338972	ESR1	0.07	**0.63**
cg01364137	GATA3	0.02	**0.70**

Same cell line name, opposite methylation state at the validated loci. MCF-7 has accumulated substantial genetic and epigenetic drift across laboratories over decades of separate passaging. Because the Roth labels encode "tumor hypomethylated at target" and Sanger's MCF-7 is methylated at those targets, any scorer whose composite contains `p_targ = P(β_tumor < 0.30)` will rank the validated positives at the *bottom* of its distribution — which is what V2.5 and `tumor_only` do. The inversion is the correct response of the scorer to the underlying biology; the labels are the component that is no longer valid on this cohort.

GSE68379 is reported as a label-transportability boundary, not a failed generalization of V2.5. Pooling it into an average across cohorts would misrepresent the result; redefining positives from GSE68379′s own β values would be circular (labels derived from the data under test). It remains in the documentation as a documented boundary case.

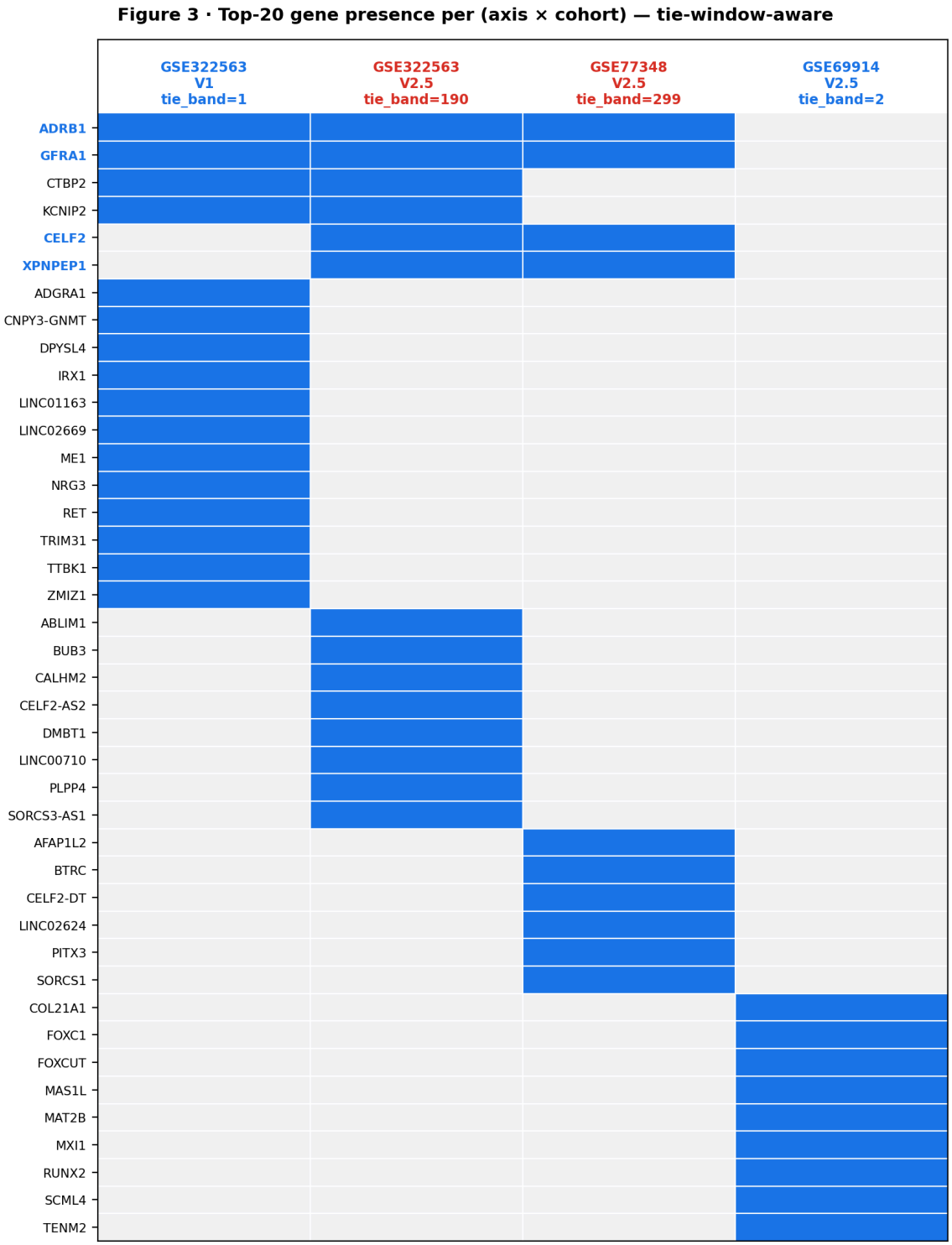

Figure 3: Figure 3 · Top-20 gene presence per (axis × cohort)

Figure 3. Per-(axis × cohort) top-20 gene presence, ordered so that the most-shared genes appear at the top. Blue column headers mark cohorts where `tie_band ≤ 2` and the top-20 is the genuine top-20 of the

score distribution; red column headers mark cohorts where `tie_band ≫ K` and the top-20 is a 20-record window inside the tied band selected by the documented `candidate_id` ascending tie-break. Bold-blue gene rows appear in BOTH cell-line V2.5 top-20 windows (GSE322563 + GSE77348). On those cohorts shared membership is window convergence, not robust ranking convergence; AUC (Fig. 2) is the stable claim at low n. V1 on GSE322563 has tie_band = 1 and samples a different (broader) genomic neighborhood — RET, ZMIZ1, DPYSL4 and other rows it surfaces are robust under that axis. The GSE69914 tissue column has tie_band = 2 and shares no genes with either cell-line column — V2.5 is cohort-specific, not returning a fixed gene list across regimes.

5.5 Top-hit annotation (tie-window-aware)

An annotation pass (nearest gene + TSS distance + feature class + CpG- island context, from UCSC `refGene` and `cpgIslandExt` hg19 tables) was run on the top-20 of each score axis × cohort. Important phrasing note: on cohorts with large tied bands at K=100, the "top-20" is a 20-record *window* inside that tied band, selected by the documented tie-break policy (`candidate_id` ascending). The membership of this window is not a robust property of the scorer in the way AUC is — a different deterministic tie-break would surface a different slice of the same band. The claims below are therefore phrased in terms of "present in the benchmark-selected top-20 window," not "the model converges on this gene."

On the two matched cell-line cohorts (GSE322563 tie_band = 190; GSE77348 tie_band = 299):

- **Present in the GSE322563 V2.5 top-20 window** (chr10 slice under `candidate_id` ascending): KCNIP2, CALHM2, SORCS3-AS1, LINC00710, CELF2 (×2), XPNPEP1, CELF2-AS2, ADRB1 (×3), ABLIM1, GFRA1 (×3), PLPP4, DMBT1, BUB3, CTBP2 (×2).
- **Present in the GSE77348 V2.5 top-20 window**: PITX3, BTRC, SORCS1, LINC02624, CELF2-DT, CELF2, XPNPEP1, ADRB1, AFAP1L2, GFRA1 (×3) (full TSV committed).
- **Shared between the two top-20 windows**: ADRB1, CELF2, GFRA1, XPNPEP1 (4 genes). This is cross-cohort convergence inside the `candidate_id`-ascending slice of the tied bands, not cross-cohort convergence of the underlying ranking. AUC is the stable claim at low `n`; top-20 membership is a window claim.

On GSE322563, V1 `final_score` has `tie_band = 1` and so its top-20 is a genuine ordering of the full candidate set, not a window slice. It surfaces different and overlapping genes:

- **GSE322563 V1 top-20** (tie_band = 1): ZMIZ1, DPYSL4, RET, LINC02669, ADRB1, KCNIP2, TRIM31 (×5), CNPY3-GNMT, TTBK1, GFRA1, IRX1, CTBP2, ADGRA1, LINC01163, ME1, NRG3. RET is included and is notable as a well-established therapeutic target in medullary thyroid cancer and RET-rearranged NSCLC; this is a robust top-20 claim under the V1 axis. Overlap with the V2.5 top-20 window on the same cohort: ADRB1, CTBP2, GFRA1, KCNIP2.

On GSE69914 (tissue, `n = 305 / 50`, tie_band = 2) the top-20 is essentially a genuine top-20:

- **GSE69914 V2.5 top-20**: RUNX2 (×5), SCML4 (×5), FOXCUT, FOXC1, MAS1L, COL21A1, MXI1, TENM2 (×3), MAT2B. β differentials are smaller (0.17–0.31) than on cell lines (0.79–0.96) — primary tissue is cell-type-heterogeneous, β regresses toward the middle. Zero gene overlap with the cell-line top-20 windows. This is the expected behavior: on a genuine top-20 (not a tied-band slice), V2.5 is cohort-specific and does not return a fixed gene list.

None of the three Roth-*Fig.-5d*-validated probes (EGFLAM T11, ESR1 T17, GATA3 T18) appear in any cohort's top-20 — they appear in the top ~1% of the scored set (rank-lifted by AUC) but not in the top- 100 slice. This is the expected behavior on cohorts with large tied bands at K = 100 and is why AUC, not P@K, is the primary endpoint.

6 · Discussion

6.1 Decision table — hierarchy of use

Three truths coexist and should not be conflated:

1. V1 `final_score` is the stable release axis (tagged `v0.4.0`).
2. V2.5 is an **experimental-on-main** probabilistic mode.
3. V2.5 is **recommended** as the probabilistic research mode for matched cell-line / paper-comparable cohorts.

The decision table below is the literal hierarchy:

intended use	axis (mode)	status	rationale
Default stable framework release	V1 `final_score`	tagged `v0.4.0`; default mode in cohort YAMLs remains `tumor_only`	Deterministic, continuous-valued score; `tie_band = 1` on every cohort tested, so P@K is never tie-break-dependent. The stable-release role is specifically about top-K interpretability, not AUC leadership — V2.5 outperforms V1 on AUC on the primary endpoint (§5.1).
Recommended probabilistic research mode, matched cell-line / paper-comparable cohorts	V2.5 (differential)	experimental-on-main, not tagged	Highest AUC at every label granularity on GSE322563 and GSE77348 (§5.1–5.2). Tie-bands reported per benchmark; P@K intervals honest. The cohort-YAML key is `probabilistic_mode: tumor_plus_differential_protection`.
Analysis-only (diagnostic)	V2 `tumor_only`	retained in the mode enum; not a discovery axis	Competitive AUC on tissue (§5.3) but `tie_band_size_at_k` consistently 6,000–12,000; top-K is not interpretable. Use only for AUC sanity checks against V2.5 / V1.
Unsupported / out-of-distribution interpretation	any axis at GSE68379	documented as §5.4 boundary case	Sanger MCF-7 epigenetic drift breaks label transportability from Roth Fig. 5d. Inverted AUC is the expected scorer response; do not pool this cohort's numbers with §5.1.

The phrase "recommended" in row 2 means: for a user running V2.5 on a cohort that matches the §5.1 use profile, we believe the reported benefit over V1 is real. It does not mean V2.5 is tagged as a release, and it does not mean V2.5 is safe to use on cohorts outside that profile without re-running the diagnostics in §5.3 and §5.4 on the new cohort.

6.2 Limitations

(Platform harmonization and catalog scope are part of the methodology proper — §4.4 — because they materially shape what the reported AUCs mean. They are cited briefly here as the methodological constraints they are.)

1. **Low-replicate tie bands.** On cohorts with `n < ramp_n = 30`, `p_trust` saturates and produces tied bands of 100s–1000s of records at the top of the probabilistic composite. `tie_band_size_at_k` and `precision_at_k_{min, max}` are reported per benchmark result so that P@K is read as an interval. The problem dissolves at `n ≥ 30` (demonstrated on GSE69914; §5.3).
2. **HM450/EPIC harmonization (see §4.4).** Reported AUCs that use GSE322563 are under the HM450-intersect path, not a native EPIC v2 scoring run. Retention of HM450 probes is 80.7% overall and 83–96% at the Roth-validated genes. A native EPIC v2 ingest would remove this caveat but is not shipped.
3. **Catalog scope (see §4.4).** AUCs are on chr5 / chr6 / chr10 candidates filtered to within ±500 bp of an HM450 probe, not on a whole-genome catalog. The three chromosomes carry the Roth Fig. 5d validated targets and provide a realistic test bed.
4. **Cell-line drift.** GSE68379 (§5.4) establishes that MCF-7 in one lab is not the same epigenetic object as MCF-7 in another. Any V2.5 target shortlist generated on a given cohort must be validated against the specific cell line that will be used in the follow-up editing assay, not against a different lab's stock of the same-named line.
5. **σ floor.** The 0.05 floor on per-side σ prevents σ_Δ from collapsing to zero at boundary β-values. The value is empirical; a formal robustness study across floor values is deferred.
6. **Top-20 membership is a window claim on low-`n` cohorts.** As §5.5 states, the cell-line top-20 lists are 20-record windows inside 190- and 299-record tied bands, selected by the documented tie-break policy. The four-gene overlap (ADRB1, CELF2, GFRA1, XPNPEP1) is convergence of the benchmark-selected window, not a robust claim about the underlying scorer ordering. AUC is the stable metric on low-`n` cohorts; top-20 membership becomes a stable metric at `n ≥ 30` (GSE69914, `tie_band = 2`).
7. **No wet-lab validation in this memo.** V2.5 is a target- prioritization tool. The claim is AUC + tie-band-aware top-K reporting on committed methylation data, not actual editing efficiency at any specific site. All named top hits are unvalidated predictions at this stage.

6.3 Next steps

In priority order, not committed to any timeline:

1. Native EPIC v2 probe annotation + catalog rebuild, to remove the HM450-intersect caveat from the Roth-comparable pipeline.
2. Annotation-pass extension: mappability, repeat overlap, and ENCODE cCRE intersection to move top-20 shortlists from "interesting genes" to "experimentally prioritized target sites".
3. A second independent-lab MCF-7 / MCF-10A EPIC cohort, if one becomes public, to establish reproducibility of the V2.5 claim at n ≥ 3 on matched cell-line pairs.
4. Wet-lab editing validation at the top V2.5 candidates on matched cell-line substrates — the terminal step and the one that actually changes the scientific claim.

---

7 · Methods — key equations

V1 `final_score`:

```
sequence_score      = PamFamily.weight
selectivity_score   = max(0, β_normal_mean − β_tumor_mean)
                    + max(0, β_normal_q25  − β_tumor_q75)
confidence_score    = EvidenceClass → {1.0, 0.7, 0.4, 0.1, 0.0}
heterogeneity_pen   = weight × max(0, β_tumor_IQR − threshold)
low_coverage_pen    = weight × max(0, (n_threshold − n_tumor) / n_threshold)
final_score         = sequence × selectivity × confidence
                    − heterogeneity_pen − low_coverage_pen
```

UNOBSERVED candidates are not penalized (verified by regression test).

V2 / V2.5 probabilistic factors:

```
p_targ = P(β_tumor < 0.30)   via method-of-moments Beta(α, β) fit,
                              fall back to piecewise-linear CDF when
                              σ² ≥ μ(1−μ). Incomplete beta via Lentz's
                              continued fraction, pure stdlib.
p_prot = P(β_normal ≥ 0.50)  via same CDF dispatch.     [V2 only]
p_diff = 1 − Φ((δ − (μ_n − μ_t)) / σ_Δ)
         where σ_Δ = √(max(σ_t, 0.05)² + max(σ_n, 0.05)²)
                           and σ_k = IQR_k / 1.349.            [V2.5]
p_trust = EvidenceClass.base × min(1, min(n_t, n_n) / 30)
         (EvidenceClass.base ∈ {0.95, 0.75, 0.45, 0.15, 0.0}).
```

Composites by mode:

```
tumor_only                          : p_targ × p_trust
tumor_plus_normal_protection        : p_targ × p_prot × p_trust
tumor_plus_differential_protection  : p_targ × p_diff × p_trust
```

Validator enforces that the stored `p_therapeutic_selectivity` matches the product implied by the declared `mode` to within `math.isclose(rel_tol=1e-9, abs_tol=1e-12)`; a second validator enforces that the differential audit fields (`p_differential_protection`, `differential_delta`) are populated iff `mode` is differential.

Benchmark ranking contract. Let the scored set be N records, each with a score `s_i` and a ground-truth label `y_i ∈ {0, 1}` (1 for positive). The benchmark sorts records by the total order

```
(s_i, c_i)   with primary  -s_i  ascending
                  secondary c_i   ascending
```

where `c_i` is the candidate_id. Let `r_1, r_2, …, r_N` be the resulting ranking (smallest `(-s, c)` first). Define:

```
top-K               := {r_1, …, r_K}
cutoff_score        := s at position K in the sort, i.e. s_{r_K}
tied_band           := {r_i : s_{r_i} = cutoff_score}          # records at the
cutoff
tie_band_size_at_k  := |tied_band|
k_band              := |tied_band ∩ top-K|                     # records in top-K at
the cutoff
drop_slots          := tie_band_size_at_k − k_band             # records at cutoff
outside top-K
band_pos            := |{r_i ∈ tied_band : y_{r_i} = 1}|
committed_pos       := |{r_i ∈ top-K \ tied_band : y_{r_i} = 1}|
```

Then:

```
P@K_observed := (committed_pos + |{r_i ∈ tied_band ∩ top-K : y_{r_i} = 1}|) / K

P@K_min      := (committed_pos + max(0, band_pos − drop_slots)) / K
P@K_max      := (committed_pos + min(band_pos, k_band))          / K
```

`P@K_min` is the smallest possible value of P@K achievable by any tie-break policy that respects the primary score ordering; it is reached by pushing as many tied-band positives as possible outside the top-K (limited by `drop_slots` available). `P@K_max` is the symmetric upper bound: pull as many tied-band positives as possible into top-K (limited by `k_band` available). By construction:

```
P@K_min ≤ P@K_observed ≤ P@K_max
```

and the interval collapses to `{P@K_observed}` exactly when `tie_band_size_at_k = 1`.

The `R@K_min` / `R@K_max` counterparts replace the denominator with `n_positives`. `AUC` is the Mann-Whitney U / (n_pos · n_neg) with ties contributing 0.5 (standard tie-handling); it does not need an interval form because under the cid-ascending tie-break it is already invariant within tied score regions.

Data and code availability

- **Code**: https://github.com/AllisonH12/thermocas9. The stable-release V1 code is tagged `v0.4.0`. The V2.5 experimental mode, the P@K-interval benchmark contract, the top-hit annotation pipeline, this manuscript, and its committed benchmark and figure artifacts all live on `main`. To cite the exact revision that produced this memo, use `git log -1 --format=%H PAPER.md` in a fresh clone — that SHA pins every behavior and artifact the text references.
- **Tests**: 215 passing under `uv run pytest -q`.
- **Cohort data**: publicly-downloadable GEO series GSE322563, GSE77348, GSE69914, GSE68379; build scripts in `scripts/build_gse*_cohort.py` produce the committed per-probe summary TSVs in `data/derived/*_cohort/`. Positives-list builder at `scripts/build_roth_positives.py` (requires the Ensembl REST `/map` endpoint for the hg38 → hg19 liftover of Roth Fig. 5d coordinates).
- **Reference data**: UCSC hg19 `refGene.txt.gz` and `cpgIslandExt.txt.gz` (fetched on demand; gitignored).
- **Benchmark artifacts**: every `BenchmarkResult` JSONL row committed under `examples/*_roth_labels/` carries `precision_at_k`, `precision_at_k_{min,max}`, `recall_at_k`, `recall_at_k_{min,max}`, `roc_auc`, `tie_band_size_at_k`, and `tie_break_policy`.
- **Annotated top-20 TSVs**: under `examples/*/top20_annotated_*.tsv`.

Reproducing the cross-cohort matrix

```bash
# Given a clone of the repo:
uv venv && source .venv/bin/activate
uv pip install -e ".[dev]"
pytest                       # 215 tests, ~1 s

# Rebuild the catalog on chr5/6/10 (~2 min):
thermocas build-catalog \
    --reference data/raw/hg19/hg19_chr5_6_10.fa \
    --pam-model config/pam_model.yaml \
    --probe-annotation data/raw/probes_hg19.tsv \
    --probe-window-bp 500 \
    --output data/derived/catalog_hg19_chr5_6_10.jsonl

# Score + benchmark one cohort (~90 s per mode on a modern laptop):
```

```
thermocas score-cohort \
    --catalog data/derived/catalog_hg19_chr5_6_10.jsonl \
    --cohort data/derived/gse322563_differential.yaml \
    --pam-model config/pam_model.yaml \
    --backend summary \
    --probe-annotation data/derived/gse322563_cohort/probes.tsv \
    --tumor-summary  data/derived/gse322563_cohort/tumor_summary.tsv \
    --normal-summary data/derived/gse322563_cohort/normal_summary.tsv \
    --probabilistic \
    --output data/derived/scored_gse322563_differential.jsonl

thermocas benchmark \
    --scored data/derived/scored_gse322563_differential.jsonl \
    --positives data/derived/positives_roth_validated.txt \
    --cohort-name GSE322563-validated-differential \
    --score-field p_therapeutic_selectivity \
    --top-k 100 --no-enforce-holdout \
    --output bench.jsonl
```

`data/derived/scored_*.jsonl` files are gitignored because they are 3.4 GB each and reproducible from the committed summary TSVs.

Citation

If this memo or the associated code is useful in your work, please cite the upstream paper:

> Roth M.O., Shu Y., Zhao Y., Trasanidou D., Hoffman R.D., et al. Molecular basis for methylation-sensitive editing by Cas9. *Nature* (2026). DOI 10.1038/s41586-026-10384-z.

For the probabilistic scoring framework specifically, cite this memo plus the code URL above.

Acknowledgements

This memo describes a framework built as an educational / research project. ThermoCas9′s biochemical characterization and the validated target sites are entirely due to Roth et al.; any errors in the re-implementation of the scoring or benchmarking are this memo's.

The deep-self-review process behind the code (documented in `V2_5_REVIEW.md`) caught a series of bugs that would have otherwise affected the headline claim: a composite-consistency validator gap on `ProbabilisticScore`, an input-order-dependent tie-break in `evaluate_ranking`, a prefix-ID tie-break bug in the top-hit annotator, and an early-termination bug in the overlap-detection loop of the gene annotator. Each of these is covered by a dedicated regression test.